

CMAKE: THE GROUND BENEATH THE CODE

What actually happens between main.cpp and a running program

WHY THIS MATTERS

Every lecture in this course has one of these

```
cmake_minimum_required(VERSION 3.20)
project(rooster)
set(CMAKE_CXX_STANDARD 23)
set(CMAKE_CXX_STANDARD_REQUIRED ON)
add_executable(rooster main.cpp)
```

You've been running `cmake -B build` and `cmake --build build` since lecture 2.2 without asking what they actually do. This lecture answers that.

WHAT CMAKE IS NOT

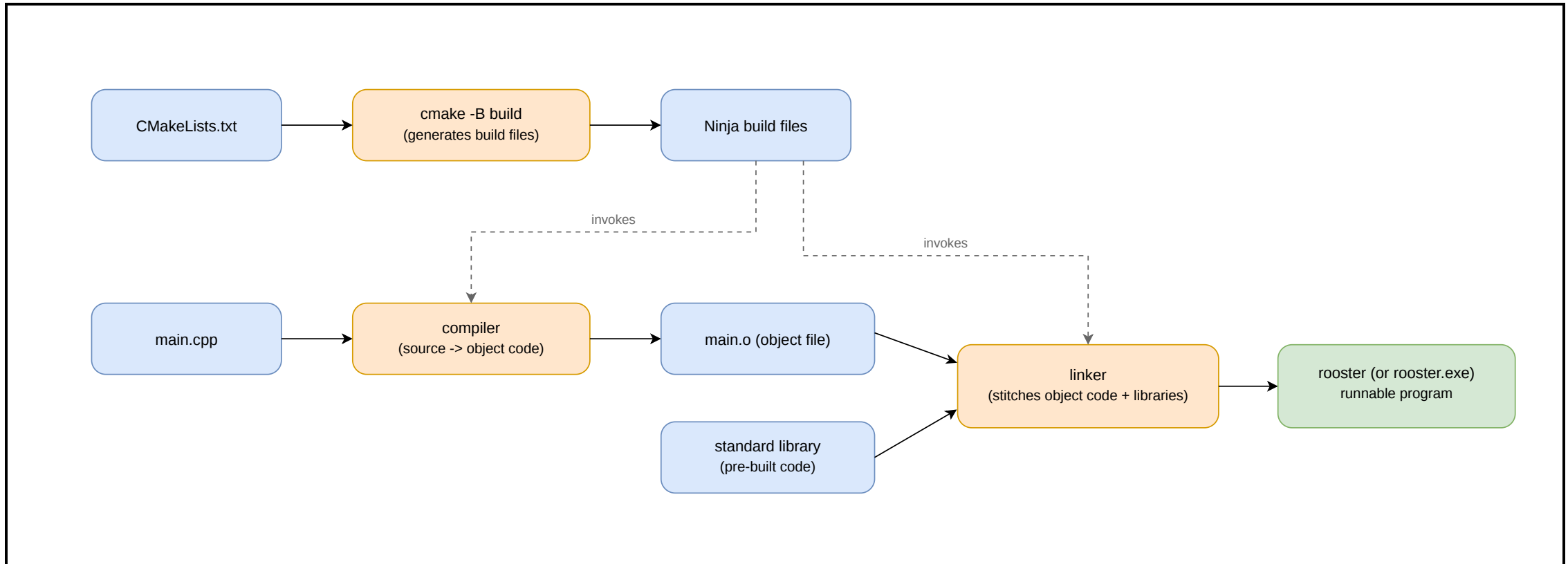
CMake doesn't compile your code

CMake is a build system generator. It never touches the compiler directly - it reads `CMakeLists.txt` and *generates* input files for a separate build tool (in this course, Ninja), which is the thing that actually invokes the compiler and linker.

```
CMakeLists.txt  --[cmake]-->  Ninja build files  --[ninja]-->  ...
```

THE FULL PIPELINE

Source code to runnable program



STEP 1: CONFIGURE

cmake -B build

```
cmake -B build -G Ninja .
```

Reads **CMakeLists.txt** and **generates** the Ninja build files inside a new **build/** folder. This step doesn't compile anything - it works out *what* needs to be compiled, in what order, with what flags, and writes that plan down.

STEP 2: BUILD

cmake --build build

```
cmake --build build
```

Runs Ninja against the plan generated in step 1. Ninja itself invokes two different programs, in order, for every source file:

1. the **compiler**
2. the **linker**

THE COMPILER'S JOB

Source code -> object code

The compiler reads one `.cpp` file at a time and translates it into **object code** - machine instructions specific to your CPU, but not yet a runnable program. On disk, this is a `.o` file (`.obj` on Windows).

```
main.cpp --[compiler]--> main.o
```

A compiler error (a typo, a missing `;`) happens at this step - your code never even reaches the linker.

THE LINKER'S JOB

Object code -> a runnable program

The linker takes your object file(s) and combines them with the standard library (pre-compiled code for things like `std::println`, `std::vector`, `std::string`) to produce one executable.

```
main.o + standard library --[linker]--> rooster (or rooster.exe)
```

A "linker error" (`undefined reference`) means the compiler understood your code fine, but something it needed wasn't found at this step.

WHERE THE NAME COMES FROM

rooster

Every `CMakeLists.txt` in this course names its executable `rooster` via `add_executable(rooster main.cpp)` - that's why `./build/rooster` (or `build\rooster.exe` on Windows) is the program you run after every build, regardless of which lecture you're in.

SAME FILE, THREE TOOLS

This is what Visual Studio and Qt Creator do for you

Neither IDE has its own separate compilation model in this course - both read this exact same `CMakeLists.txt`, and both are really just running `cmake -B build` / `cmake --build build` behind a Run/Debug button. Docker (2.6) runs the identical two commands by hand, inside a container. One build description, three ways to trigger it.